

HIDE: Short IDs for Robust and Anonymous Linking of Users Across Multiple Sessions in Small HCI Experiments

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ABSTRACT

Record linking is needed to analyze observations across multiple sessions. However, recent privacy legislature such as the General Data Protection Regulations (GDPR) restricts the storage of information that identify individuals. Obtaining permissions to store information about individuals can be bureaucratic and time-consuming. Anonymous schemes such as self-generated ids and machine generated ids have been proposed. However, self-generated linking ids demand effort from the participants, while machine assisted schemes typically generate long and incomprehensible ids. Consequently, there is a risk that students and researchers will limit their research to single-session experiments to avoid privacy issues. To simplify the administration of small multi-session experiments, the HIDE procedure is proposed for generating short human readable ids for linking participants across multiple sessions while maintaining anonymity and being robust to input errors. The approach is different from previous approaches in that the goal is to minimize the length of the linking ids. First, the procedure converts the participant's name into a phonetic representation. Next, this phonetic representation is hashed, and a truncated snippet of the hash is used as the linking id. HIDE is initialized by searching for a salt (a random data added to the hash input) that minimizes the id lengths. Experiments show that the procedure is capable of coding small experiments with 20 participants using two digits, and experiments of around 200 participants with four digits. An implementation of the procedure has been made available through a simple web interface. It is hoped that the procedure can help students and HCI researchers collect more comprehensive data by following participants over time, while protecting their privacy.

CCS CONCEPTS

• **CCS Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

KEYWORDS

Longitudinal studies, Multi session experiments, Record linking, Soundex, Hashing, Privacy, GDPR

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1 INTRODUCTION

To get a deeper understanding of new interactive technologies, it is necessary to follow participants over time as they gain practice. Longitudinal experiments, pre-test/post-test experiments, and other experimental designs involving multiple sessions are common in HCI [23, 36, 54], for example observing how participants learn new text entry methods over time [30, 42, 56]. Traditionally, the observations from the various sessions were administered using linking tables, where participants' names were assigned unique ids (running numbers). These ids were used to link observations from different sessions. Each individual datafile could be stripped of any information that revealed the identity of the participant. However, the existence of a linking table poses a risk to the participant's privacy if it is leaked. To overcome such problems, numerous procedures have been proposed including self-generated ids [57] and machine generated ids [38]. It is a goal to keep sessions short, and the process of establishing self-generated ids may divert the attention away from the experiment. Machine-based coding schemes [38] usually generate long and incomprehensible ids that do not contribute to participants' trust and perceived privacy. Long ids may seem like overkill in HCI experiments which often have around 12 participants [4]. There is also an ongoing debate about the ability of such methods to preserve participants' anonymity [35]. It has been found that participants do not always trust anonymization procedures [1]. Participants' decision about whether to give informed consent is balanced around the social sensitivity of the research questions and the common good that may result from the particular research [1].

Many research institutions have tightened their policies with the introduction of the General Data Protection Regulations (GDPR) to protect the privacy of individuals. Privacy is especially critical in accessibility experiments that involves disabled participants [2, 10, 44]. Researchers need to apply for permission to store research data with information that identify participants such as their name, phone number, e-mail address and IP-address. Applying for such permissions can be time-consuming, bureaucratic, and daunting for students who are conducting their first HCI experiment. Students may be particularly discouraged from conducting longitudinal experiments as courses typically run across a limited period, making it impractical to collect formal permissions. Consequently,

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researchers may decide to settle for single session experiments to avoid the administrative burden of acquiring permissions. Decisions to omit the collection of data across multiple sessions may therefore impact the quality of the results negatively for certain types of research questions.

A simple procedure is therefore proposed for generating short human-readable linking ids to help researchers and students easily administer multi-session experiments while preserving the privacy of the participants. The procedure assumes that the experimenter initially has a list of names of potential participants. This list is used to search for an experiment-specific salt that gives minimum-length ids that uniquely discriminate the participants. During the experiment, the session is labelled with the unique id generated using the participant's name, the salt, and id length. The anonymity of the id is ensured by the principle of k -anonymity [47] where the short ids give multiple hits when applied to unrelated names. For example, with three-digit ids there will be statistically 1,000 hits per id with a brute-force attack using a list of one million names.

2 RELATED WORK

Longitudinal studies are sometimes used in HCI as it is often relevant to observe user experience over time [23, 36, 54]. Several frameworks for longitudinal observations within the field of HCI [20, 53] have not explicitly addressed anonymous linking of data.

However, the problem of linking of anonymous data from several sources has received a vast amount of attention within the medical domain, among researchers working on register data/microdata, and more recently among data scientists [16, 49, 52]. It has even been claimed that this problem is associated with most publications within computer science [26]. Fault tolerant linking has received particular attention as records may be incomplete and contain errors. Incorrect record linkage in research may result in biased results [17].

Early approaches simply converted the names of the individuals using the soundex phonetic coding function [18, 33]. Linking was performed by comparing the phonetic codes. The phonetic coding step made the procedure tolerant to certain types of input errors such as spelling mistakes. The soundex function is lossy and irreversible, and originally it was considered to provide sufficient privacy [33]. The anonymity of soundex codes is probably contested by most today as these codes provide enough information to identify individuals in certain contexts. A more recent soundex proposal added fake records to obfuscate the data [22].

Thoben et al. [48] explored anonymous record linkage using control numbers generated from one-way (lossy) functions. They investigated six control number coding schemes including (a) surname, date of birth, and sex, (b) first character of the surname, first name, birthdate, and sex, (c) soundex coded names, (d) sum of the ASCII codes of characters in the surname and family name string, date of birth, and sex, (e) three first characters of the surname, first name, month and year of birth, and sex, and (f) a variation on (e) without the month of birth.

Later approaches applied irreversible hash functions to the resulting phonetic representations [3, 37, 38]. Records were then linked by comparing hash codes. Simply encrypting records is not considered adequate [3] as encryption keys may get lost or be subject

to cryptographic attacks. Although hash functions are irreversible, it is possible to conduct dictionary attacks, i.e., if we encode the list of all names from a phonebook, the hash code would provide us with a reliable confirmation that a given person is in a dataset. Hash codes are therefore often salted, i.e., some string is added to the value before hashing [38]. Linking schemes based on hashing without phonetic coding have also been proposed [21]. Weber et al. [55] first reduced the information content in the data by using the two first characters from each of the first name and surname and the date of birth when creating the hash codes.

Phonetic methods have been criticized for proving more false positives than other methods [5, 13, 40, 51]. Most of the research effort during the last decade has therefore been on probabilistic record linking procedures based on Bloom filters [38, 45]. In short, the Bloom filter approach involves extracting consecutive character pairs of the names into bigrams, then these bigrams are converted using several different hashing functions and mapped into a binary vector (typically 1,000 bits). Record linking is performed by comparing the bit vectors. It is thus possible to get partial matches, and the method is robust to several types of input errors.

Bloom filter methods have been criticized for being vulnerable to cryptanalysis attacks [6, 11, 25, 39, 45] and may therefore not provide sufficient privacy to participants. One suggested improvement is to apply salting to bigram encodings [35]. Bloom filter approaches are often mentioned in the context of large data integration and are known for being slow [27]. Performance can be improved by splitting the data into blocks that are processed separately [34].

Hash codes are usually long. The well-known SHA and MD hashing functions with 160 and 128 bits respectively have been used to link data [38] and several Bloom filter proposals employ 1000 bits. If representing such hash codes in human readable format a 1,000-bit, Bloom filter coding requires a string of 250 hex digits, a 160-bit SHA hash requires 40 hex digits, and a 128-bit MD hash requires 32 hex digits (each hex-digit represents a group of four bits). Many of the algorithms that rely on long hash codes, such as Bloom filters, are intended for machine linking of large public records with information about millions of individuals. We argue that such ids are less suitable for manual administration of smaller experiments as strings of 32 hex digits (or more) would typically appear incomprehensible to participants. Clearly, such long strings of seemingly random sequences exceed the capacity of short-term memory (7 ± 2) [32].

Self-generated identification ids constitute a manual approach to anonymous record linking procedures. In this approach each participant first answers a brief questionnaire. The questionnaire responses are used to generate unique ids that are used to tag the data from the sessions. Each time the participant attends a session the same self-generated id is created by the participant, allowing comparisons to be made across multiple sessions. Yet, the ids are intended to ensure the anonymity of participants. Many self-generated coding schemes have been proposed [8, 9, 15, 24, 50]. Yurek et al. [57] suggested a six-character id built from the answers to four questions: (a) the first letter of the mother's first name, (b) the number of older brothers (two digits), (c) month of birth (two digits), and (d) first name of middle name. Another example is Lippe et al.'s [28] id comprising (a) age, (b) first letters of middle name, the mother's first name, father's first name, grandmother's first name,

own surname, number of older brothers, sisters, mother's siblings, modulus 13 of the first character of the first name, month of birth, and first letter of birthplace.

Self-generated ids may contain errors and Levenshtein distances have been explored as one remedy [46]. It has been found that self-generated ids with sensitive data can be vulnerable to attacks if the data are not sufficiently diverse [29, 31]. Glanti et al. [14] studied errors associated with self-generated ids and concluded that self-generated ids are inefficient for longitudinal studies.

3 THE HIDE PROCEDURE

In a typical multi-session experiment, one may use a linking table to analyze data across different sessions. A linking table typically assigns running numbers to the participants. Such running numbers do not reveal any personal information about the participants, but if the linking table is leaked the participants are no longer anonymous. The objective is thus to eliminate the linking table to protect the anonymity of the participants. An algorithm is needed that can assign such running (as close as possible to consecutive) numbers without knowledge about the list of participants.

Formally the problem can be defined as assigning N unique ids to N participants from a superset of M unique integer ids. This is known as perfect hashing [41]. In addition, the goal is to minimize the range of M , and in the optimal case $N = M$. This optimal case is known as minimal perfect hashing [7, 12, 41]. Minimal perfect hash functions require stored information to map the keys, while the proposed HIDE procedure does not require any such stored information. The effectiveness of the resulting ids can be defined as $E = N/M$, where an encoding is optimal if $E = 1$.

A novel aspect of the proposed approach involves an initialization step where the full list of names is used to search for a salt that minimizes the id lengths. The list of names is then discarded. To retrieve the linking id for a participant, the given salt is provided to the encoding procedure together with the participant's name.

The encoding comprises four steps: sorting the parts of each name alphabetically, converting the name to a phonetic representation, hashing, and truncating. The phonetic coding step and the hashing step are similar to those already used in several approaches [3, 37, 38] while the sorting step and the truncation step are unique to the HIDE procedure. The purpose of sorting the name parts alphabetically is to ensure that the procedure is robust to the name input order. For example, "Norman Donald" should give the same id as "Donald Norman".

Next, each part of the sorted name is converted to a phonetic representation using a modified soundex algorithm. The soundex algorithm preserves the first letter of a word, removes all vowels including the quasi-vowel *w*. The remaining consonants are mapped to a more coarse-grained phonetic representation using digit codes, namely 1: {b, f, p, v}, 2: {c, g, j, k, q, s, x, z}, 3: {d, t}, 4: {l}, 5: {m, n} and 6: {r}. Repeated digits are removed if they appear consecutively in the original name. For example, "Donald Norman" would be coded as "D543 N655". Unlike the original soundex algorithm which only retained the three first digits, HIDE retains all the digits.

This phonetic simplification is intended to serve two purposes. First, it allows for more robust input with mistakes as it ignores incorrect vowels and is tolerant to incorrect transcription, for example,

mistaking a *t* for a *d* and incorrect spelling of single/double letters. Second, the phonetic simplification may reduce the information content in the name, thereby leading to shorter ids. As the robustness of soundex has already been studied thoroughly [18, 33] it was not addressed herein.

The main purpose of the hashing step is to map the names uniformly across the set of possible ids. Moreover, the one-way hash function obfuscates the identities. The webtool reported herein uses the simple djb2 hash function while the experiments rely on the object hash function built into java. Note that a salt is added to the phonetic representation prior to hashing.

The final step involves truncating the hash code by retaining the d least significant digits. For example, the hash code "57836278" would give an id of 278 if the ids are to comprise $d = 3$ digits. The purpose of the truncation step is to ensure anonymity. A hash code can be used to uniquely identify a participant, while a truncated hash code is not unique to a specific participant. Short ids provide a higher degree of anonymity than longer ids.

Clearly, there is a risk that two different names may end up with the same id. The procedure therefore involves an initialization step applied to the full list of participants. First, the list of names is checked for duplicates; next, the list of names in soundex representation is checked for duplicates to ensure that two names do not result in identical phonetic representations. Then, an exhaustive search is performed for a salt that leads to the fewest number of digits that uniquely discriminates all the names. This salt is added to each phonetic representation of the names before hashing. The 3,000 most frequent English words were used as potential salts. It is assumed that a simple word would be easier to memorize or transcribe than a salt comprising an arbitrary sequence of characters.

4 EXPERIMENTAL EVALUATION

To demonstrate the viability of the proposed linking procedure, a list of 105,026 researchers worldwide based on Scopus data from a study by Ioannidis et al. [19], later adopted by [43], was used. The author list was cleaned as follows. First, names leading to 1,551 duplicate entries were identified and removed giving a list of 103,474 entries. Next, 5,911 names leading to duplicate phonetic representations were removed. The resulting list comprised 97,573 unique names.

4.1 Id Lengths

First, the performance of the procedure was assessed for sample sizes of 10 to 200 participants in steps of 10. In context of HCI this range would cover small to large experiments. For each sample size the specified number of names were randomly drawn from the list of 97,573 names and input to the procedure. This step was repeated 100 times for each participant sample size to get representative results across a range of randomly drawn names. The minimum, maximum, and mean number of id digits obtained are plotted against participant sample size in Figures 1 and 2

Figure 1 shows the results obtained using the HIDE procedure. The HIDE procedure successfully encoded 200 participants using 4 digits ($E = 0.2$) in the worst cases, 80 participants with 3 digits ($E = 0.08$) in the worst cases, and 20 participants with 2 digits ($E = 0.2$). In the best case 10 participants were encoded with 1 digit ($E = 1$), up

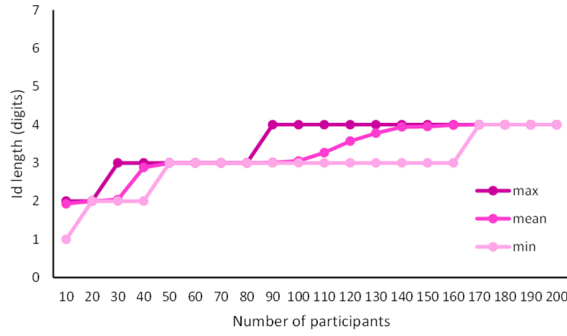


Figure 1: Id lengths of the HIDE procedure as a function of participant sample size in the best case, mean case, and worst case.

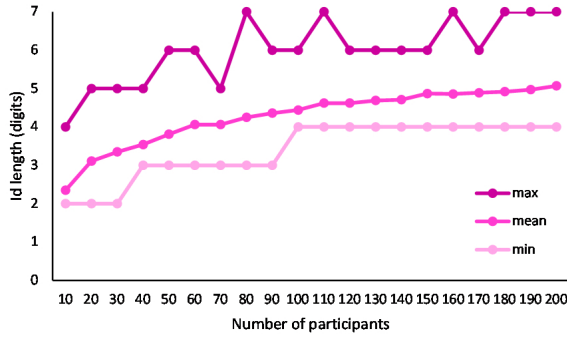


Figure 2: ID lengths without optimized salt as a function of participant sample size in the best case, mean case, and worst case.

to 40 participants with 2 digits ($E = 0.4$), and up to 160 participants with 3 digits ($E = 0.16$). The ability to anonymously encode 200 participants is sufficient for many types of HCI experiments which often have as little as 12 participants [4].

The results obtained without the phonetic coding step (direct hashing of names) are nearly identical to those presented in Figure 1 (and therefore not included). The sum of differences shows that HIDE only exhibited a negligible 0.03 overall improvement over the direct hashing of the names. Clearly, the phonetic coding step did not lead to the expected improved coding efficiency in the resulting hashes.

Figure 2 shows the results obtained when applying the procedure without the optimized salt. Clearly, the ids are longer when the optimized salt is not used. In fact, in several instances the worst-case yields ids with as much as 7 digits ($E < 0.00001$). With 7 digits privacy of participants probably cannot be preserved. In the mean case 4 digits are needed with 60 participants ($E = 0.006$), and 3 digits are needed with 30 participants ($E = 0.03$). These results demonstrate the importance of the optimized salt to effectively map the participants into a smaller range of id values.

The detailed evaluation was limited to 200 participants due to the computational effort required. To assess the limits of HIDE,

Table 1: Coding capacity for larger participant sample sizes.

Participant sample size	Id lengths (digits)
400	5
800	5
1,600	6
3,200	6
6,400	7
12,800	20

Table 2: Effects of the 3000 salts based on a list of 163 names.

digit length	Frequency
4	1,003
5	1,402
6	570
7	22

snapshots of how the procedure performs with very large experiments are shown in Table 1. With 5 digits it was possible to encode 800 participants. In practice, 5 digits are probably the largest number of digits that should be used as it can map 100,000 unique entries. With 6 digits it is one million possible mappings, which means that there is a larger risk being able to confirm that someone is a participant in an experiment. Table 1 shows that the coding scheme collapsed with 12,800 participants.

Figure 2 confirms that the optimized salts gave shorter ids. To assess the effects of the salt, the results of a search for the optimal salt with 163 names were recorded as an example case (see Table 2). Most salts resulted in 5-digit ids, while about one third of the salts gave the minimum id length of 4 digits. Several salts resulted in 5, 6, and even 7-digit ids.

4.2 Anonymity

To illustrate the anonymity provided by the proposed short ids, the full list of 97,573 names was fed back to the coding for one case from each sample size from 10 to 200. Figure 3 shows the distribution of names that maps to each of the possible ids (both valid and invalid ids). With two-digit ids there were approximately 1,000 names from the list that mapped to each of the entries. With three-digit ids there were about 100 names mapped to each id, and with four-digit ids there were a mean of 10 names mapped to each id with a minimum of one name per id. Clearly, the scale of the statistics plotted in Figure 3 is linked to the list size. The size of this example is comparable to the population of Antigua and Barbuda (97,929 people in 2021).

5 DISCUSSION

With the list of more than 100,000 author names there was a 5.7% chance of phonetic collisions. With small experiments, which are the most common within HCI [4], the chance of phonetic collisions is low. One strategy for handling sound collisions such as “Lena Hansen”, “Lene Hanson”, and “Line Hansson” could be for the researcher to add an extra description using some mnemonic aid to

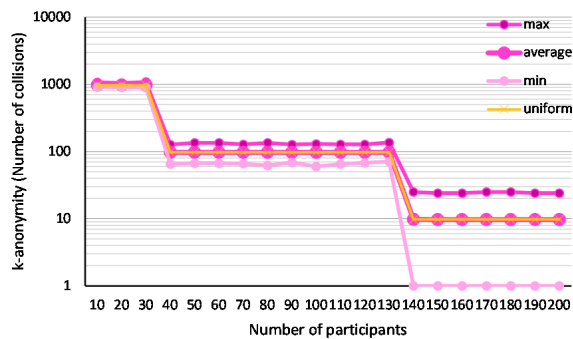


Figure 3: A log-linear plot of k -anonymity as a function of participant sample size in the best case, average case, worst case and uniformly distributed (expected). The results are based on 97,573 names.

help discriminate between the names. Examples include descriptions that give meaning and associations to the researcher when interacting with the participant “Lena Hansen decaffeinated”, “Lene Hanson parking permit”, and “Line Hansson administrator”. The experimenter would have to remember these special cases. Alternatively, the phonetic coding step can be omitted if there are collisions in the phonetic representation.

Obviously, there is a theoretical risk of hash collisions. However, the experimental evaluations showed that the salt prevents collisions in practice. The HIDE works with languages transcribed using the Latin alphabet. For other languages, such as Chinese, other schemes are needed in place of soundex. Alternatively, the phonetic coding step may be omitted by hashing the names directly, thereby sacrificing the tolerance to input errors.

Sorting the name parts alphabetically before encoding ensures that the parts of a name can be input in an arbitrary order. If one part of the name is missing (e.g., middle name or initial), the ids will not match, and the procedure will fail. Methods based on bigrams (including Bloom filters) are more tolerant to missing name parts.

The examples outlined herein use the participants’ full names. Alternatively, one may also use the participants’ first names or nicknames if these are unique. Small HCI experiments with less than 100 participants will typically yield ids with 2-3 digits. Most people are familiar with three-digit numbers, and these may therefore be perceived as less mystical and more trustworthy than long hash codes.

New participants may be added gradually to small samples (10-20 participants) through snowball sampling if slightly longer ids are chosen beforehand (3 or 4 digits). When recruiting a new participant, the experimenter needs to check if the new name leads to a collision. If so, the name must be altered with an easily memoizable change. Yet, multi-session experiments typically necessitate more systematic recruitment of participants than single session experiments. Clearly, there is a tradeoff between the inflexibility of upfront recruitment of participants versus the administrative burden of acquiring formal permissions to store linking tables, or manually administering name collisions.

A JavaScript implementation of the procedure is made available through a simple web-interface running locally in the browser at

<https://www.cs.oslomet.no/~frodes/HIDE/> as well as an experiment implementation in java and the name lists used at <https://github.com/frode-sandnes/HIDE>.

6 CONCLUSION

The HIDE procedure for assigning short human-readable, anonymous, and error-tolerant linking ids to participants was presented. Experiments demonstrated the practicality of the approach. An implementation of the coding procedure running in the browser is available. It is hoped that the procedure will make it easier to gain participants’ trust in the preservation of their anonymity. In addition, it is hoped that the procedure will encourage more HCI students and researchers to observe users over time.

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